

17. Which one of the following statements concerning forced vibrations and resonance is correct?
- A** An oscillating body that is not resonating will return to its natural frequency when the forcing vibration is removed.
 - B** At resonance, the displacement of the oscillating body is 180° out of phase with the forcing vibration.
 - C** A pendulum with a dense bob is more heavily damped than one with a less dense bob of the same size.
 - D** Resonance can only occur in mechanical systems.

Answer: A

- B:** Incorrect. At steady state, displacement of forced oscillation is in-phase with the driving force.
If the driver is applied out-of-phase with the initial oscillation of the body, there will be a transient state of motion during which the body changes its motion to coincide with that of the driver.
- C:** This is a common misconception, mass $\neq$ damping.
- D:** Forced oscillations occurs in both mechanical and non-mechanical (e.g. electrical circuit such as inductor-capacitor circuit) systems.